

TECHNICAL & COMMERCIAL PROPOSAL

185.4 kWp Rooftop Solar PV System

Kampala Marriott Hotel – Mezzanine Roof

Prepared for	Kampala Marriott Hotel, Kampala, Uganda
Prepared by	Nile Sun Power Limited
Proposal ref.	P-KMH-PV-2026-001
Date	5 November 2026
Validity	30 days

1. Executive Summary

Nile Sun Power Limited is pleased to propose a turnkey 185.4 kWp grid-tied solar PV system on the mezzanine roof of the Kampala Marriott Hotel. The system generates clean electricity during the day that is used entirely by the hotel, reducing purchases from the grid. It operates with zero export and needs no battery storage.

185.4 kWp Solar capacity	**274,362 kWh** Year-1 generation	**USD 19,212** Year-1 savings	**6.1 years** Simple payback
USD 121,115 Turnkey investment	**16.6 %** Project IRR (25 yrs)	**USD 71,221** NPV @ 10 %	**USD 569,043** 25-year net savings

Why this project makes sense

- **Fast payback:** the investment is recovered in about 6.1 years, and the system then keeps saving for more than 18 more years.
- **Low-cost energy:** solar electricity costs about USD 0.052/kWh over the system life, below the grid tariff of about USD 0.070/kWh – and grid tariffs are expected to rise.
- **Every unit is used:** the hotel's daytime demand is far above the solar output, so 100 % of the energy offsets grid purchases. No export, no battery.
- **Sustainability:** about 6.5 GWh of clean energy over 25 years, supporting the hotel's environmental commitments.
- **Turnkey delivery:** design, supply, installation, commissioning and handover by a single contractor, in about 12–14 weeks.

2. Site & Energy Profile

Location	Kampala Marriott Hotel, Kampala, Uganda (0°18'11.34"N, 32°35'13.76"E)
Installation area	Mezzanine-level flat concrete roof – architect's solar zone (drawing D18102-DD-A-104)
Hotel consumption	≈ 1,000,000 kWh per month (≈ 12 GWh per year)
Grid supply	UEDCL, time-of-use tariff; main LV switchboard with 1000 A incomer
Solar resource	Equatorial site – stable output all year, ≈ 1,480 kWh per kWp per year



Figure 1 – Proposed PV array on the mezzanine roof (indicative visual)

3. Proposed Solution

The system is a grid-tied rooftop PV plant feeding the hotel's main LV switchboard. Solar power is always used first; the grid supplies the balance. A zero-export controller measures the hotel's grid import and limits solar output so that no power flows back to the utility.

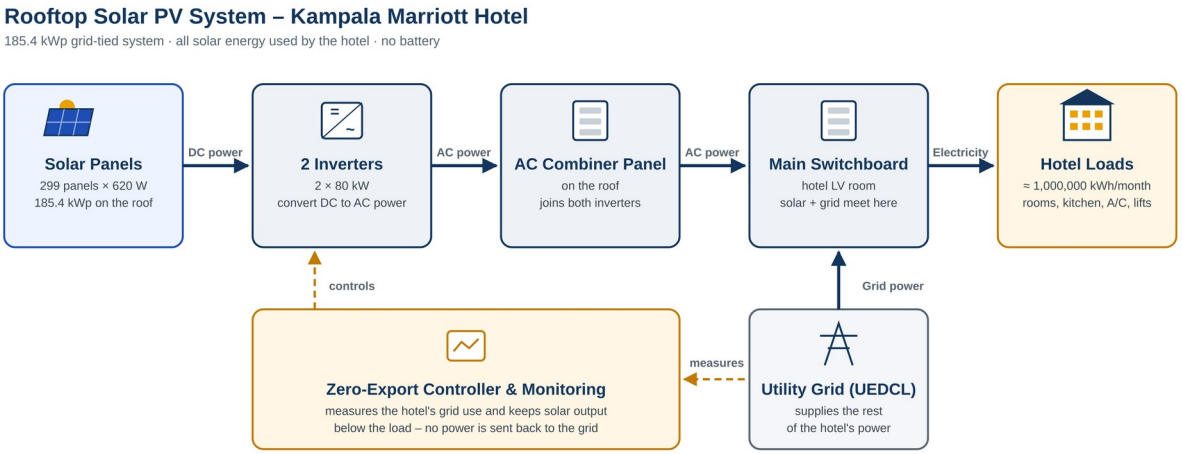


Figure 2 – System block diagram

Component	Description	Size / Quantity
Solar panels	620 Wp bifacial N-type TOPCon, Tier-1, portrait, 10° tilt	299 nos = 185.4 kWp
Inverters	Three-phase string inverters, 1100 V, multi-MPPT	2 × 80 kW = 160 kW AC
Mounting structure	Ballasted aluminium system, no roof penetration	7 tables (6 × 2P, 1 × 1P)
AC combiner panel	Joins both inverters; MCCBs, surge protection, roof isolator	1 panel, 250 A
Grid connection	New 250 A way on the existing main LV switchboard	1 feeder
Monitoring & control	Smart data logger, remote monitoring, zero-export control	1 set
Earthing & protection	Array bonding, connection to building earth and lightning protection	1 lot

4. Energy Generation

The system is expected to generate about 274,362 kWh in its first year – about 22,900 kWh per month or 750 kWh per day. Because Kampala lies on the equator, output is stable through the year: the best months (January–March) are only about 20 % above the lowest (June–July).

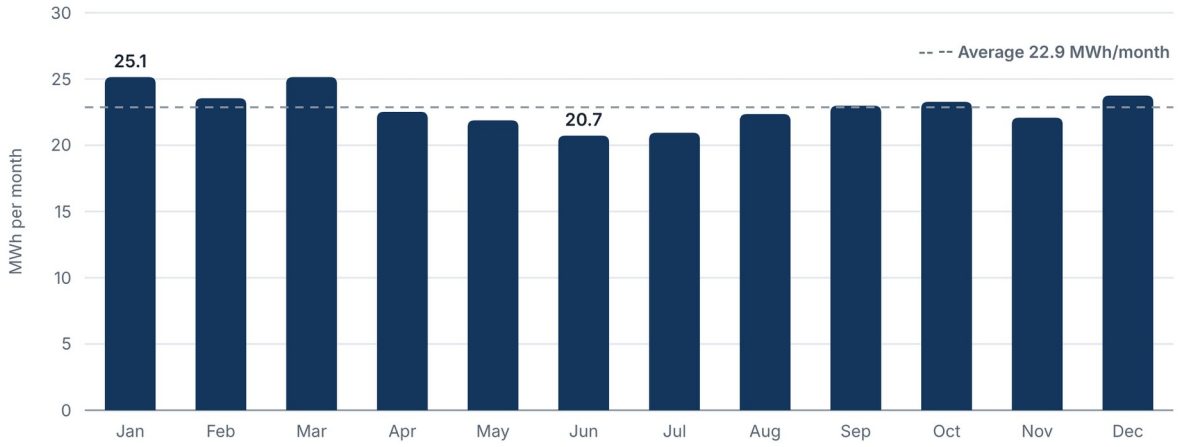


Figure 3 – Estimated monthly generation, year 1 (MWh)

Month	GHI (kWh/m²/day)	Days	Generation (kWh)	kWh per day	Share of year
Jan	5.4	31	25,143	811	9.2 %
Feb	5.6	28	23,551	841	8.6 %
Mar	5.4	31	25,143	811	9.2 %
Apr	5.0	30	22,529	751	8.2 %
May	4.7	31	21,884	706	8.0 %
Jun	4.6	30	20,727	691	7.6 %
Jul	4.5	31	20,952	676	7.6 %
Aug	4.8	31	22,349	721	8.1 %
Sep	5.1	30	22,980	766	8.4 %
Oct	5.0	31	23,280	751	8.5 %
Nov	4.9	30	22,079	736	8.0 %
Dec	5.1	31	23,746	766	8.7 %
Total		**365**	**274,363**	**752**	**100 %**

Almost all generation (≈ 98.6 %) falls in the shoulder tariff period (06:00–18:00), when the hotel's demand is high, so every kWh directly reduces grid purchases. Panel output degrades slowly (about 0.4 % per year); total generation over 25 years is about 6,477 MWh.

5. Financial Analysis

5.1 Key assumptions

Turnkey investment (excl. VAT)	USD 121,115
Avoided grid tariff (year 1)	USD 0.070/kWh, escalating 2 % per year
Generation, year 1 / degradation	274,362 kWh / 1 % in year 1, then 0.4 % per year
Inverter replacement	USD 9,600 in year 12
Discount rate / analysis period	10 % / 25 years

5.2 Results

Year-1 savings	USD 19,212
Simple payback	6.1 years
Project IRR (25 years)	16.6 %
Net present value @ 10 %	USD 71,221
Levelised cost of solar energy	USD 0.052/kWh
25-year net savings (after inverter replacement)	USD 569,043
Net gain after recovering the investment	USD 447,928
Return on investment (25 years) / year-1 return	370 % (4.7× the investment) / 15.9 %

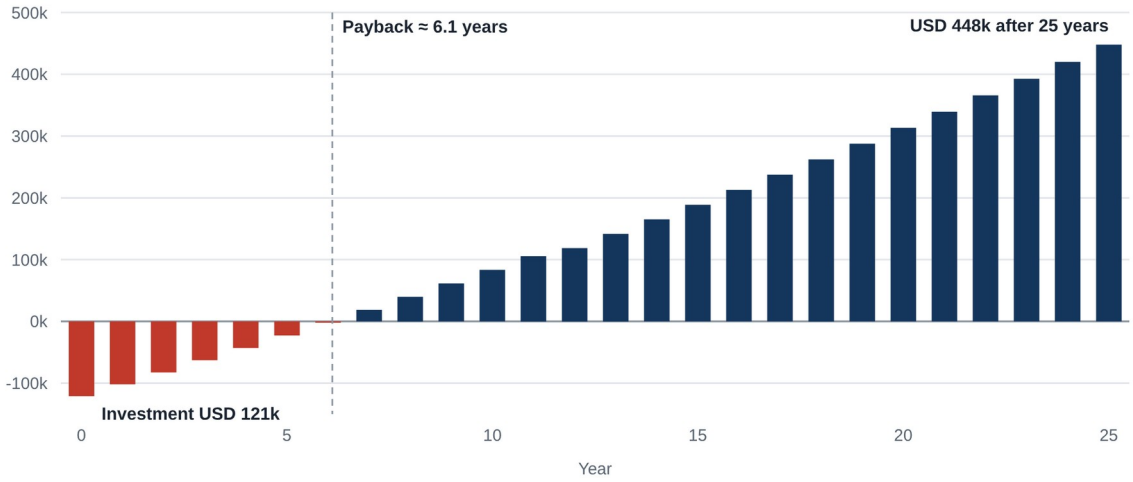


Figure 4 – Cumulative cash flow (USD)

5.3 Cash flow (selected years, USD)

Year	Generation (kWh)	Tariff (USD/kWh)	Savings	Net cash flow	Cumulative
0				(121,115)	(121,115)
1	274,362	0.070	19,212	19,212	(101,903)
2	270,532	0.071	19,322	19,322	(82,581)
3	269,450	0.073	19,630	19,630	(62,951)
4	268,372	0.074	19,942	19,942	(43,009)
5	267,299	0.076	20,260	20,260	(22,749)
6	266,230	0.077	20,582	20,582	(2,167)
7	265,165	0.079	20,910	20,910	18,744
8	264,104	0.080	21,243	21,243	39,987
10	261,996	0.084	21,925	21,925	83,492
12	259,904	0.087	22,628	13,028	118,795
15	256,797	0.092	23,726	23,726	188,864
20	251,702	0.102	25,676	25,676	313,284
25	246,708	0.113	27,786	27,786	447,928

Year 12 includes the USD 9,600 inverter replacement. Values in brackets are negative (investment not yet recovered).

5.4 Tariff sensitivity

Avoided tariff (USD/kWh)	Year-1 savings (USD)	Payback (years)	IRR	NPV @ 10 % (USD)
0.05	13,718	8.4	11.5 %	15,349
0.06	16,462	7.1	14.1 %	43,253
0.07	**19,205**	**6.1**	**16.6 %**	**71,158**
0.08	21,949	5.4	19.0 %	99,062
0.09	24,693	4.8	21.4 %	126,967
0.10	27,436	4.3	23.8 %	154,871

The project remains attractive across a wide tariff range. The break-even tariff (NPV = 0 at 10 %) is about USD 0.044/kWh – roughly 36 % below today's tariff – which gives a strong safety margin.

Financial figures are indicative and exclude O&M costs, which can be offered separately (see Section 7).

6. Scope of Works & Project Schedule

6.1 Included

- Detailed engineering: layout, single-line diagram, string design, cable sizing and protection.
- Supply of all equipment: PV modules, inverters, mounting structure, DC and AC systems, monitoring and zero-export control.
- Transport, customs clearing and delivery to site in Kampala.
- Installation, cabling, earthing and connection to the main LV switchboard.
- Testing, commissioning, as-built documentation, operator training and handover.

6.2 Client responsibilities & exclusions

- Roof access, secure storage area, and power and water during installation.
- A spare 250 A way on the main LV switchboard and access to the LV room.
- Excluded: roof structural strengthening or waterproofing, battery storage, utility or permit fees, VAT and statutory levies.

6.3 Indicative schedule

Phase	Timing	Key output
Detailed design & approvals	Weeks 1–3	Approved drawings, structural sign-off
Procurement & shipping	Weeks 2–10	Equipment delivered to site
Installation	Weeks 10–13	Structure, modules, cabling, panels
Testing & commissioning	Weeks 13–14	System energised, handover

7. Warranties & After-Sales Support

Item	Warranty
PV modules	12-year product warranty; 30-year linear power warranty
Inverters	5 years (extendable)
Mounting structure	10 years
Workmanship	2 years from commissioning

Optional operation & maintenance (O&M) services – panel cleaning, periodic inspections, remote performance monitoring and reporting – can be provided under a separate annual agreement.

8. Commercial Offer – Quotation

Quotation No.	Q-KMH-PV-2026-001	Date	05-11-2026
Revision	A	Valid until	05-12-2026 (30 days)
Client	Kampala Marriott Hotel	Supplier	Nile Sun Power Limited
Address	Kampala, Uganda	Contact	Mahil Fernando
Project	185.4 kWp rooftop PV, mezzanine roof	Mobile / Email	+94 773 55 96 82 · mahil@windforce.lk

No.	Scope of supply & works	Qty	Unit	Total Price (USD)
1	PV modules – 620 Wp bifacial N-type TOPCon (Tier-1)	299	nos	USD 121,115
2	String inverters – 80 kW, 1100 V, multi-MPPT	2	nos	
3	Ballasted aluminium mounting structure, 10° tilt	1	lot	
4	DC system – PV cable, connectors, string isolators, cable tray	1	lot	
5	AC system – combiner panel, MCCBs, cabling to main LV switchboard	1	lot	
6	Smart data logger, remote monitoring & zero-export control	1	set	
7	Earthing & lightning protection integration	1	lot	
8	Installation, testing & commissioning	1	lot	
9	Transport, logistics & customs clearing to site	1	lot	
TOTAL TURNKEY EPC PRICE (excluding VAT and other applicable taxes)				**USD 121,115**

Equivalent to USD 0.65 per Wp. Amount in words: One hundred twenty-one thousand one hundred fifteen US Dollars only.

Price basis	Turnkey EPC – delivered, installed and commissioned at site, Kampala. Currency: USD.
Payment terms	30 % advance with purchase order · 60 % on delivery of main equipment to site · 10 % on commissioning
Taxes	Prices exclude VAT, withholding tax and other statutory levies, charged as applicable.
Delivery	Equipment 8–10 weeks from PO and advance; installation and commissioning ≈ 4 weeks after delivery.
Warranties	Modules 12-yr product / 30-yr power · inverters 5 yrs · structure 10 yrs · workmanship 2 yrs
Notes	Final quantities subject to detailed engineering and site survey. Exclusions as listed in Section 6.2.

For Nile Sun Power Limited	Accepted by Kampala Marriott Hotel
Signature / Name / Date	Signature / Name / Date

9. Next Steps

1. Review of this proposal and clarification meeting with the hotel's engineering team.
2. Site survey: roof condition, LV switchboard, cable route and inverter location.
3. Confirmation of the tariff from recent utility bills and structural review of the roof.
4. Purchase order and advance payment – project start.

We thank the Kampala Marriott Hotel for the opportunity and look forward to delivering this project together.

Contact: Mahil Fernando, Nile Sun Power Limited · +94 773 55 96 82 · mahil@windforce.lk